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Real-Time AI/NLP Risk Engine & Tactical Index Rebalancer

An explainable intelligence pipeline for headline-driven portfolio decisions

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Track: AI/NLP Risk Engine + Module A



PROTOTYPE ARCHITECTURE • OCTOBER 2026

Turn unstructured headlines into controlled decisions

THE CHALLENGE

News is fast, noisy and ambiguous. Generic sentiment misses financial context; unconstrained LLM outputs can be costly and hard to audit.

THE RISK

A headline can change perceived credit or business risk before traditional workflows reflect it. Passive weights do not react to new information.

THE APPROACH

Convert text into structured, traceable signals, then feed those signals into a constrained tactical allocation layer.

PIPELINE PRINCIPLES

Domain-aware

Financial lexicon + negation handling

Explainable

Store rationale and source attribution

Risk-controlled

Long-only weights with concentration limits

Design goal: make each portfolio change traceable to a source event and a reproducible rule.

A modular pipeline from ingestion to allocation



Separation of concerns: extraction produces signals; the rebalancer independently enforces portfolio constraints.

Domain-aware signals, bounded decisions

01 FINANCIAL NLP CALIBRATION

Use the Loughran–McDonald lexicon as a domain baseline; apply negation scope rules and event categories to reduce obvious polarity errors.

02 EXPLAINABILITY BY DESIGN

Persist source text, extracted ticker, sentiment, event type, impact score and rationale so a decision can be reviewed later.

03 TIME-DECAYED SIGNALS

Weight recent events more heavily. For half-life h, use $\lambda = \ln(2)/h$ and decay each signal by $e^{(-\lambda \Delta t)}$.

04 PORTFOLIO GUARDRAILS

Long-only allocation, 2% floor and 25% cap are enforced after tilting; weights are normalized to 100% when feasible.

Performance claims such as “<2 ms per headline” should be reported only after measuring the implemented pipeline under a stated test setup.

Measure signal quality and portfolio value separately

NLP / SYSTEM METRICS

F1 / precision

Event classification — Evaluate on a manually labelled test set

p50 / p95 latency

Per-headline runtime — Report hardware, batch size and warm-up

Cache hit rate

Repeated-text efficiency — Track API calls avoided and failure rate

PORTFOLIO METRICS

Excess return

vs. equal-weight benchmark — Same dates, universe and rebalance schedule

Max drawdown

Peak-to-trough loss — Include a clear out-of-sample period

Turnover + costs

Implementation realism — Include spread, fees and slippage

Evidence status: populate these fields with measured results. Positive alpha and sub-2 ms latency are hypotheses until a reproducible backtest/benchmark confirms them.

A decision-support layer for analysts and index teams

EARLY-WARNING WORKFLOW

Surface potential credit, regulatory and earnings risks for analyst review—without claiming to replace formal ratings.

ACTIVE INDEX RESEARCH

Test whether headline-derived signals improve risk-adjusted outcomes versus a transparent benchmark.

AUDITABILITY

Link signal, source headline, scoring rationale and resulting weight change in one reviewable trail.

POTENTIAL WORKFLOW

Detect
New headline or filing

→

Assess
Structured signal + rationale

→

Review
Constrained allocation suggestion

Positioning: decision support for research and surveillance, not an autonomous trading or credit-rating system.

Build credibility through explicit assumptions

CURRENT SCOPE / LIMITATIONS

- Mock constituent universe and simplified execution assumptions
- Headline quality, entity linking and sarcasm can affect signals
- Backtest results can suffer from look-ahead and survivorship bias
- Transaction costs and market impact need modeling

NEXT ITERATIONS

- Add FinBERT/transformer embeddings for nuanced financial text
- Use asynchronous workers and retry/dead-letter handling
- Add spreads, fees, slippage and turnover penalties
- Evaluate walk-forward out-of-sample performance

VALIDATION CHECKLIST

- **Signal quality**
- **Latency**
- **Portfolio**
- **Governance**

Thank you